Additional Question Bank — Solved (95 questions)

البنك الإضافي — 95 سؤالاً مع الإجابات ومصادرها

Source: the two extra banks sent by the student: a solved copy and an unsolved practice copy of the **same 95 questions** (the file's cover says 85, but it contains 95; I checked that both copies have the identical questions). The questions cover Ch1–7, 14 and 16 and do not repeat the doctor's exam.

- Answers and the per-question source (Studocu, Quizlet, Vaia…) are exactly as printed in the solved file. **I did not open those links**; I only checked the answers against the lecture notes and found no disagreement.
- The unsolved copy is covered by the interactive quiz, where answers stay hidden until you press Check.

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Ch1: Computer Organization & Evolution

True / False — (5) صح / خطأ

1 **Organizational attributes include hardware details transparent to the programmer, control signals, interfaces, and memory technology.**

Answer: **True**

Source in the file: Studocu CH01 Testbank — https://www.studocu.com/row/document/jamaa%D8%A9-aayn-shms/history-of-architecture-iii/ch01-testbank/62838876

2 **The IAS computer (completed 1952) is the prototype of all subsequent general-purpose computers based on Von Neumann's stored-program concept.**

Answer: **True**

Source in the file: Quizlet CEA201 Unit 2 — /https://quizlet.com/60387317/cea201-unit-2-flash-cards

3 **In the IAS computer, the Program Counter (PC) contains the address of the next instruction-pair to be fetched from memory.**

Answer: **True**

Source in the file: Quizlet CPSC 440 — /https://quizlet.com/572083696/cpsc-440-exam-1-flash-cards

4 **In 3rd generation (ICs), the building blocks of a digital computer are Gates, Memory cells, and Interconnections.**

Answer: **True**

Source in the file: Studocu COA Generations — https://www.studocu.com/row/document/mount-kenya-university/computer-architecture/computer-architecture-and-organization-key-concepts-and-generations/131776823

5 **Moving from Traditional IT to SaaS makes the architecture less complex, lower upfront cost, more scalable, and less customizable.**

Answer: **True**

Source in the file: Studocu Ch1 Notes — https://www.studocu.com/row/document/pokhara-university/computer-engineering/chapter-1-introduction-to-computer-organization-and-architecture/135543937

Multiple Choice — (11) اختيار من متعدد

1 **The software/hardware boundary in the hierarchy of computer architecture is:**

A. Compiler
B. Operating System
C. Instruction Set Architecture (ISA) ✓
D. Digital Design

Answer: **C. Instruction Set Architecture (ISA)**

Source in the file: Studocu Ch1 Notes — https://www.studocu.com/row/document/pokhara-university/computer-engineering/chapter-1-introduction-to-computer-organization-and-architecture/135543937

2 **Which is NOT one of the three characteristics of Von Neumann architecture?**

A. Four main subsystems (Memory, ALU, CU, I/O)
B. Program stored in memory
C. Instructions executed in parallel by default ✓
D. Sequential execution

Answer: **C. Instructions executed in parallel by default**

Source in the file: Studocu Von Neumann — https://www.studocu.com/en-gb/document/university-of-greenwich/computer-systems-architectures/computer-system-architecture-lecture-notes/16604747

3 **The four basic functions a computer performs are data processing, data storage, data movement, and:**

A. Compilation
B. Control ✓
C. Virtualization
D. Indexing

Answer: **B. Control**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

4 **In the IAS computer memory format, each 40-bit instruction word contains:**

A. One 40-bit instruction
B. Two 20-bit instructions (each: 8-bit opcode + 12-bit address) ✓
C. Four 10-bit instructions
D. Two 16-bit instructions

Answer: **B. Two 20-bit instructions (each: 8-bit opcode + 12-bit address)**

Source in the file: Studocu COMP204 Q&A — https://www.studocu.com/en-za/document/university-of-kwazulu-natal/computer-organisation-and-architecture/comp204-test-1-complete-past-paper-qa-bank-memorandum/170994578

5 **In the IAS computer, the _____ temporarily holds the right-hand instruction from a word in memory.**

A. MBR
B. IR
C. IBR (Instruction Buffer Register) ✓
D. MAR

Answer: **C. IBR (Instruction Buffer Register)**

Source in the file: Quizlet Ch1 Flashcards — https://quizlet.com/368982593/ch1-computer-organization-and-/architecture-flash-cards

6 **In the IAS computer, the _____ contains the 8-bit opcode instruction being executed.**

A. MBR
B. IBR
C. IR (Instruction Register) ✓
D. MAR

Answer: **C. IR (Instruction Register)**

Source in the file: Quizlet CEA Ch2 MC — /https://quizlet.com/vn/245353731/cea-chapter-2-mc-flash-cards

7 **In the 2nd generation (IBM 7094), a 'data channel' is:**

A. A cache line in ALU
B. An independent I/O module with its own processor and instruction set ✓
C. A vacuum tube unit
D. An optical disk

Answer: **B. An independent I/O module with its own processor and instruction set**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

8 **Which chip (1971) was the first to contain all components of a CPU on a single chip (birth of microprocessor)?**

A. Intel 8008
B. Intel 8080
C. Intel 4004 ✓
D. IBM 7094

Answer: **C. Intel 4004**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

9 **The Intel 8008 (1972) was the first 8-bit microprocessor, while the _____ (1974) was the first general-purpose microprocessor.**

A. Intel 4004
B. Intel 8080 ✓
C. Intel 8086
D. Pentium

Answer: **B. Intel 8080**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

10 **A single chip containing the processor, ROM, RAM, clock, and I/O control unit ('computer on a chip') is a:**

A. Multiplexor
B. Microcontroller ✓
C. Data channel
D. Discrete component

Answer: **B. Microcontroller**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

11 **In Cloud Computing, which service model lets customers deploy their own applications while the CSP manages everything from Application Framework down to Networking?**

A. IaaS
B. PaaS (Platform as a Service) ✓
C. SaaS
D. Traditional IT

Answer: **B. PaaS (Platform as a Service)**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

Ch2: Performance Issues in Computer Systems

True / False — (2) صح / خطأ

1 **Pipelining reduces the execution time of an individual instruction.**

Answer: **False**

Source in the file: Studocu Comp Arch Guide — https://www.studocu.com/en-us/document/harvard-university/computer-science/comp-arch-cs101-final-exam-study-guide-pipelining-cache/163776812

2 **The Multicore strategy uses two or more simpler processors on a chip rather than one complex processor, with 2 or 3 levels of cache on chip.**

Answer: **True**

Source in the file: Quizlet Ch1 Stallings — https://quizlet.com/417090536/chapter-1-basic-concepts-and-/computer-evolution-flash-cards

Multiple Choice — (4) اختيار من متعدد

1 **What distinguishes Superpipelined from Superscalar execution?**

A. Superpipelined divides stages into granular sub-stages for higher clock frequency; Superscalar issues multiple instructions per cycle via multiple execution units ✓

B. They are identical

C. Superpipelined uses multiple ALUs simultaneously

D. Superscalar cannot use pipelines

Answer: **A. Superpipelined divides stages into granular sub-stages for higher clock frequency; Superscalar issues multiple instructions per cycle via multiple execution units**

Source in the file: Studocu Ch16 Superscalar — https://www.studocu.com/ph/document/centro-escolar-university/computer-engineering/chapter-16-superscalar-execution-and-ilp-insights/154254841

2 **Using branch prediction and data flow analysis to execute instructions ahead of their actual appearance in the program, holding results in temporary locations, is:**

A. Superpipelining

B. Speculative execution ✓

C. DMA

D. Interleaving

Answer: **B. Speculative execution**

Source in the file: Medium C. Kekula — https://medium.com/@ckekula/understanding-computer-performance-branch-prediction-31efe31cf162

3 **In _____ analysis, the processor analyzes which instructions are dependent on each other's results to create an optimized schedule.**

A. Branch prediction

B. Data flow ✓

C. Cache mapping

D. Bus arbitration

Answer: **B. Data flow**

Source in the file: Medium C. Kekula — https://medium.com/@ckekula/understanding-computer-performance-branch-prediction-31efe31cf162

4 **Which approach helps overcome the performance imbalance between processor and DRAM speeds?**

A. Make DRAMs wider rather than deeper

B. Reduce memory access frequency via efficient caches

C. Increase interconnect bandwidth with high-speed bus hierarchies

D. All of the above ✓

Answer: **D. All of the above**

Source in the file: Quizlet CEA201 Unit 2 — /https://quizlet.com/60387317/cea201-unit-2-flash-cards

Ch3: Top-Level View of Computer Function & Interconnection

True / False — (1) صح / خطأ

1 **On a shared bus, if two devices transmit during the same time period, their signals overlap and become garbled.**

Answer: **True**

Source in the file: COA Garage Bus — http://coagarage.blogspot.com/p/bus.html

Multiple Choice — (13) اختيار من متعدد

1 **Which register specifies the address in memory for the next read or write?**

A. MBR
B. MAR (Memory Address Register) ✓
C. I/OAR
D. I/OBR

Answer: **B. MAR (Memory Address Register)**

Source in the file: Quizlet CEA Chap 3+4 — /https://quizlet.com/36766615/cea-chap-34-flash-cards

2 **The _____ contains the data to be written into memory or receives the data read from memory.**

A. MAR
B. PC
C. MBR (Memory Buffer Register) ✓
D. I/OAR

Answer: **C. MBR (Memory Buffer Register)**

Source in the file: Quizlet CEA201 Ch3 — /https://quizlet.com/vn/550164187/cea201-flash-cards

3 **I/OAR specifies a particular I/O device, whereas _____ is used for the exchange of data between an I/ O module and the CPU.**

A. MBR
B. MAR
C. I/OBR (I/O Buffer Register) ✓
D. IR

Answer: **C. I/OBR (I/O Buffer Register)**

Source in the file: Studocu Ch3 Notes — https://www.studocu.com/row/document/south-east-european-university/computer-architecture/chapter-iii-a-top-level-view-of-computer-function-and-interconnection/44778266

4 **The four action categories of computer function are Processor-memory, Processor-I/O, Data processing, and:**

A. Caching
B. Control (altering execution sequence) ✓
C. Pipelining
D. Bus grant

Answer: **B. Control (altering execution sequence)**

Source in the file: Medium BICS1302 — https://medium.com/bics1302-stury-reflection/introduction-to-computer-organisation-week-2-sect-4-2412181-haddif-asjad-8e0286efae4d

5 **In the Instruction Cycle State Diagram, which state analyzes the instruction to determine the operation and operands?**

A. Instruction fetch (if)
B. Instruction operation decoding (iod) ✓
C. Operand address calculation (oac)
D. Data operation (do)

Answer: **B. Instruction operation decoding (iod)**

Source in the file: Blogspot / Medium — https://maniruzzaman-akash.blogspot.com/2017/04/list-and-briefly-define-possible-states.html

6 **In the Instruction Cycle State Diagram, which state determines the address of an operand in memory or I/O?**

A. iac
B. Operand address calculation (oac) ✓
C. Operand fetch (of)
D. Operand store (os)

Answer: **B. Operand address calculation (oac)**

Source in the file: Blogspot / Medium — https://maniruzzaman-akash.blogspot.com/2017/04/list-and-briefly-define-possible-states.html

7 **The four classes of interrupts are Program, Timer, I/O, and:**

A. Cache miss
B. Hardware failure (power failure / memory parity error) ✓
C. Branch prediction
D. Bus grant

Answer: **B. Hardware failure (power failure / memory parity error)**

Source in the file: Quizlet CEA Chap 3+4 — /https://quizlet.com/36766615/cea-chap-34-flash-cards

8 **Arithmetic overflow, division by zero, or executing an illegal instruction generates a _____ interrupt.**

A. Timer
B. Program ✓
C. I/O
D. Hardware failure

Answer: **B. Program**

Source in the file: Quizlet CEA201 Final — https://quizlet.com/vn/518426888/cea201-final-exam-version-3-/flash-cards

9 **A _____ interrupt is generated by an internal processor timer to allow the OS to perform functions regularly.**

A. Program
B. Timer ✓
C. I/O
D. Hardware failure

Answer: **B. Timer**

Source in the file: Quizlet CEA Chap 3+4 — /https://quizlet.com/36766615/cea-chap-34-flash-cards

10 **Which mechanism allows I/O modules to read/write directly to memory without tying up the processor?**

A. Programmed I/O
B. Direct Memory Access (DMA) ✓
C. ISR
D. Associative Mapping

Answer: **B. Direct Memory Access (DMA)**

Source in the file: Quizlet CEA201 Final — https://quizlet.com/vn/518077332/cea201-final-exam-flash-/cards

11 **The width of the _____ bus determines how many bits can be transferred at a time and is a key factor in system performance.**

A. Control
B. Address
C. Data ✓
D. Interrupt

Answer: **C. Data**

Source in the file: Studocu System Bus — https://www.studocu.com/en-gb/document/wigston-college/computer-science/data-address-and-control-bus-explained/118115708

12 **Which control bus line indicates that data have been accepted from or placed on the bus?**

A. Bus grant
B. Bus request
C. Transfer ACK ✓
D. Interrupt ACK

Answer: **C. Transfer ACK**

Source in the file: COA Garage Bus — http://coagarage.blogspot.com/p/bus.html

13 **Which control line indicates a module needs control of the bus, whereas 'Bus grant' indicates it has been granted control?**

A. Memory write
B. Bus request ✓
C. Transfer ACK
D. Reset

Answer: **B. Bus request**

Source in the file: COA Garage Bus — http://coagarage.blogspot.com/p/bus.html

Ch4: Computer Memory System Overview

True / False — (2) صح / خطأ

1 **Internal memory is directly accessible by the processor, while external memory is accessed via an I/ O module.**

Answer: **True**

Source in the file: Quizlet CEA201 Unit 4 — /https://quizlet.com/60391758/cea201-unit-4-flash-cards

2 **Cache is NOT a form of internal memory.**

Answer: **False**

Source in the file: Quizlet CEA201 Unit 4 — /https://quizlet.com/60391758/cea201-unit-4-flash-cards

Multiple Choice — (5) اختيار من متعدد

1 **'Memory is organized into records and access must be made in a specific linear sequence' describes:**

A. Sequential access (Tape) ✓
B. Direct access (Disk)
C. Random access (RAM)
D. Associative access (Cache)

Answer: **A. Sequential access (Tape)**

Source in the file: Quizlet CEA201 Unit 4 multi — /https://quizlet.com/130331190/cea201-unit-4-multi-flash-cards

2 **Individual blocks have a unique address based on physical location and use a shared read-write mechanism in:**

A. Associative access
B. Direct access (Magnetic Disk) ✓
C. Random access
D. Sequential access

Answer: **B. Direct access (Magnetic Disk)**

Source in the file: Quizlet CEA201 Unit 4 multi — /https://quizlet.com/130331190/cea201-unit-4-multi-flash-cards

3 **Going down the memory hierarchy (Registers -> Cache -> Main Memory -> Disk -> Tape) results in:**

A. Decreasing cost/bit, increasing capacity, increasing access time ✓
B. Increasing cost/bit and decreasing capacity
C. Decreasing access time
D. Higher transfer rate

Answer: **A. Decreasing cost/bit, increasing capacity, increasing access time**

Source in the file: Quizlet Ch4 COA — https://quizlet.com/sa/166830635/ch4-computer-architecture-/flash-cards

4 **A portion of main memory used as a buffer to hold data temporarily for disk transfers is called a:**

A. Disk cache ✓
B. Register file
C. Virtual address
D. TLB

Answer: **A. Disk cache**

Source in the file: Quizlet CEA201 Unit 4 multi — /https://quizlet.com/130331190/cea201-unit-4-multi-flash-cards

5 **Registers are managed by the _____, Cache by _____, and Main Memory by _____.**

A. Compiler; Processor hardware; Operating system (OS) ✓
B. OS; Compiler; User
C. Hardware; OS; Compiler
D. User; OS; Hardware

Answer: **A. Compiler; Processor hardware; Operating system (OS)**

Source in the file: ManaTutor Memory — https://www.manatutor.com/show.php/list-and-explain-the-different-characteristics-of-memory-system?id=209

Ch5: Cache Memory Principles

True / False — (2) صح / خطأ

1 **A Logical Cache sits between the processor and MMU (uses virtual addresses), whereas a Physical Cache sits between the MMU and main memory (uses physical addresses).**

Answer: **True**

Source in the file: Quizlet Ch4 Cache — https://quizlet.com/309564861/chapter-4-computer-memory-/system-overview-cache-memory-flash-cards

2 **In a two-level cache, L1 is internal (on-chip) and usually split (instruction/data), while L2 is external/higher level and unified.**

Answer: **True**

Source in the file: Quizlet CEA201 Unit 4 — /https://quizlet.com/130331190/cea201-unit-4-multi-flash-cards

Multiple Choice — (7) اختيار من متعدد

1 **Each cache line includes a _____ that identifies which main-memory block is currently stored in that line.**

A. Hit bit
B. Tag ✓
C. Page table
D. Sector

Answer: **B. Tag**

Source in the file: Quizlet CEA201 Unit 4 multi — /https://quizlet.com/130331190/cea201-unit-4-multi-flash-cards

2 **_____ is the simplest mapping technique and maps each block of main memory into only one possible cache line.**

A. Direct mapping ✓
B. Associative mapping
C. Set-associative mapping
D. Virtual mapping

Answer: **A. Direct mapping**

Source in the file: Quizlet CEA201 Unit 4 multi — /https://quizlet.com/130331190/cea201-unit-4-multi-flash-cards

3 **In _____ mapping, cache lines are grouped into v sets of k lines (m = v x k), and each memory block maps to one unique set (any line in that set).**

A. Direct
B. Fully associative
C. Set-associative ✓
D. Sequential

Answer: **C. Set-associative**

Source in the file: CliffsNotes Cache — https://www.cliffsnotes.com/study-notes/22311893

4 **Which mapping technique needs NO replacement algorithm because each block has only one fixed cache line?**

A. Associative
B. Set-associative
C. Direct mapping ✓
D. All of them

Answer: **C. Direct mapping**

Source in the file: Studocu Cache Principles — https://www.studocu.com/in/document/trident-academy-of-technology/computer-science/cache-memory-principles-understanding-design-and-functionality/139286339

5 **Which replacement algorithm replaces the block in the set that has been in the cache longest with NO reference to it, and is the most popular?**

A. FIFO
B. LFU
C. Least Recently Used (LRU) ✓
D. Random

Answer: **C. Least Recently Used (LRU)**

Source in the file: Quizlet Ch4 Cache — https://quizlet.com/309564861/chapter-4-computer-memory-/system-overview-cache-memory-flash-cards

6 **Which replacement algorithm replaces the block with the fewest references using a counter on each line?**

A. LRU
B. FIFO
C. Least Frequently Used (LFU) ✓
D. Random

Answer: **C. Least Frequently Used (LFU)**

Source in the file: Studocu Tutorial 4 — https://www.studocu.com/my/document/universiti-malaya/computer-system-architecture/tutorial-4/61952186

7 **With the _____ write policy, updates are made only in the cache (minimizing memory writes), making parts of main memory invalid.**

A. Write-through
B. Write-back ✓
C. Write-once
D. Read-only

Answer: **B. Write-back**

Source in the file: Studocu Cache Concepts — https://www.studocu.com/en-us/document/stanford-university/advanced-topics-in-computer-architecture-polymorphic-processors/organization-and-architecture-9/61474755

Ch6: Internal Memory

True / False — (3) صح / خطأ

1 **DRAM is simpler, smaller, denser, and cheaper than SRAM (used for main memory), whereas SRAM is faster and used for cache.**

Answer: **True**

Source in the file: Ques10 SRAM vs DRAM — /https://www.ques10.com/p/10049/compare-sram-dram-1

2 **The three methods for adding bits to detect single-bit errors are Parity Method, Cyclic Redundancy Check (CRC), and Hamming Code.**

Answer: **True**

Source in the file: TutorialsPoint Error Correction — https://www.tutorialspoint.com/article/what-is-error-correction

3 **Cache DRAM (CDRAM) integrates a small SRAM cache (16 Kb) onto a generic DRAM chip to serve as a true cache or serial buffer.**

Answer: **True**

Source in the file: Quizlet CEA201 Unit 5 — /https://quizlet.com/60391972/cea201-unit-5-flash-cards

Multiple Choice — (10) اختيار من متعدد

1 **Which semiconductor memory is VOLATILE (loses data without power) and erased electrically at the byte level?**

A. ROM
B. PROM
C. EPROM
D. RAM (DRAM & SRAM) ✓

Answer: **D. RAM (DRAM & SRAM)**

Source in the file: Quizlet CEA201 Unit 5 — /https://quizlet.com/60391972/cea201-unit-5-flash-cards

2 **In _____, binary values are stored using flip-flop logic-gate configurations (6 transistors), whereas DRAM uses 1 capacitor + 1 transistor.**

A. ROM
B. SRAM ✓
C. DRAM
D. Flash

Answer: **B. SRAM**

Source in the file: Quizlet CEA201 Unit 5 multi — /https://quizlet.com/130331280/cea201-unit-5-multi-flash-cards

3 **A _____ contains a permanent pattern of data wired in during fabrication, cannot be changed, and if one bit is wrong the whole batch is thrown away.**

A. RAM
B. SRAM
C. ROM ✓
D. EEPROM

Answer: **C. ROM**

Source in the file: Quizlet CEA201 Unit 5 multi — /https://quizlet.com/130331280/cea201-unit-5-multi-flash-cards

4 **Which nonvolatile memory is cheaper than ROM, written only ONCE electrically by the customer after fabrication, and needs special equipment?**

A. PROM ✓
B. EPROM
C. EEPROM
D. DRAM

Answer: **A. PROM**

Source in the file: Medium Internal Memory — https://medium.com/@ckekula/understanding-computer-internal-memory-ad894025dce0

5 **How is an EPROM erased?**

A. Electrically (byte-level)
B. Electrically (block-level)
C. UV light (chip-level) ✓
D. Cannot be erased

Answer: **C. UV light (chip-level)**

Source in the file: Ebrary EPROM — https://ebrary.net/206226/computer_science/eprom_erasable_prom

6 **With _____, a section of memory cells is erased electrically in a single action ('flash' / block-level), and it is intermediate between EPROM and EEPROM in cost.**

A. Flash memory ✓
B. SDRAM
C. DRAM
D. PROM

Answer: **A. Flash memory**

Source in the file: Quizlet CEA201 Unit 5 multi — /https://quizlet.com/130331280/cea201-unit-5-multi-flash-cards

7 **In _____ memory, DRAM chips are grouped into K banks that can service K read/write requests simultaneously.**

A. Associative
B. Interleaved ✓
C. Virtual
D. Optical

Answer: **B. Interleaved**

Source in the file: Quizlet CEA201 Unit 5 — /https://quizlet.com/60391972/cea201-unit-5-flash-cards

8 **_____ are permanent physical defects (stuck at 0/1) from manufacturing/environmental abuse, whereas _____ are random nondestructive events from power supply problems.**

A. Hard failures; Soft errors ✓
B. Soft errors; Hard failures
C. Parity; CRC
D. Misses; Hits

Answer: **A. Hard failures; Soft errors**

Source in the file: Quizlet CEA201 Unit 5 multi — /https://quizlet.com/130331280/cea201-unit-5-multi-flash-cards

9 **_____ moves data in and out under the control of the system clock without imposing wait states.**

A. Asynchronous DRAM
B. SDRAM (Synchronous DRAM) ✓
C. EPROM
D. ROM

Answer: **B. SDRAM (Synchronous DRAM)**

Source in the file: Quizlet CEA201 Unit 5 multi — /https://quizlet.com/130331280/cea201-unit-5-multi-flash-cards

10 **_____ sends data twice per clock cycle using both the rising and falling edges of the clock.**

A. CDRAM
B. SDRAM
C. DDR-SDRAM ✓
D. SRAM

Answer: **C. DDR-SDRAM**

Source in the file: Quizlet CEA201 Unit 5 multi — /https://quizlet.com/130331280/cea201-unit-5-multi-flash-cards

Ch7: External Memory

True / False — (1) صح / خطأ

1 **The internal components of an SSD architecture are Interface, Controller, Addressing, Data buffer/ cache, Error correction, and Flash memory components.**

Answer: **True**

Source in the file: Studocu Storage Devices — https://www.studocu.com/row/document/kenyatta-university/computer-programming/form-1-notes-storage-devices/168051058

Multiple Choice — (6) اختيار من متعدد

1 **Reduction in surface defects, better stiffness, and better shock/damage resistance are benefits of using _____ on a magnetic disk.**

A. Magnetic heads
B. Aluminum alloy
C. Glass substrate ✓
D. Optical laser

Answer: **C. Glass substrate**

Source in the file: Quizlet CEA201 Unit 6 multi — /https://quizlet.com/133384791/cea201-unit-6-multi-flash-cards

2 **Each disk surface consists of concentric rings called _____, and each of those consists of _____ separated by gaps.**

A. Sectors; Tracks
B. Tracks; Sectors ✓
C. Spindles; Platters
D. Cylinders; Arms

Answer: **B. Tracks; Sectors**

Source in the file: Quizlet COMP 7300 Ch6 — https://quizlet.com/844097451/comp-7300-chapter-6-external-/memory-flash-cards

3 **A _____ disk has one read/write head per track, whereas a _____ disk has one read/write head per surface on a radial arm.**

A. Movable-head; Fixed-head
B. Fixed-head; Movable-head ✓
C. Winchester; Floppy
D. Single-sided; Double-sided

Answer: **B. Fixed-head; Movable-head**

Source in the file: Quizlet COMP 7300 Ch6 — https://quizlet.com/844097451/comp-7300-chapter-6-external-/memory-flash-cards

4 **Which head mechanism operates in sealed assemblies free of contaminants, closer to the disk surface, rising on spinning air pressure?**

A. Contact (floppy)
B. Fixed gap
C. Aerodynamic gap (Winchester) ✓
D. Optical laser

Answer: **C. Aerodynamic gap (Winchester)**

Source in the file: Quizlet COMP 7300 Ch6 — https://quizlet.com/844097451/comp-7300-chapter-6-external-/memory-flash-cards

5 **Compared to HDDs (up to 55g operating / 350g non-operating), SSDs resist shocks up to:**

A. 100g
B. 350g
C. 1500g ✓
D. 10g

Answer: **C. 1500g**

Source in the file: Dell / Studocu Ch7 Notes — https://www.dell.com/support/contents/en-us/article/product-support/self-support-knowledgebase/data-storage-backup-and-recovery/support-for-hard-disk-drive

6 **What is the correct capacity order of Optical Discs from smallest to largest?**

A. DVD < CD < Blu-ray
B. CD (700 MB) < DVD (4.7-8.5 GB) < Blu-ray (25-128 GB) ✓
C. Blu-ray < DVD < CD
D. CD < Blu-ray < DVD

Answer: **B. CD (700 MB) < DVD (4.7-8.5 GB) < Blu-ray (25-128 GB)**

Source in the file: Studocu Storage Devices — https://www.studocu.com/row/document/kenyatta-university/computer-programming/form-1-notes-storage-devices/168051058

Ch14: Instruction Sets: Addressing Modes & Formats

True / False — (1) صح / خطأ

1 **Variable-length instructions provide variations compactly, but increase processor complexity because the next instruction length is unknown.**

Answer: **True**

Source in the file: Studocu Ch14 Review — https://www.studocu.com/row/document/south-east-european-university/computer-architecture/chapter-xiv-instruction-sets-addressing-modes-and-formats/44778232

Multiple Choice — (9) اختيار من متعدد

1 **In _____ addressing (Operand = A), the operand is inside the instruction itself, used to initialize registers to a constant value.**

A. Immediate ✓
B. Direct
C. Indirect
D. Stack

Answer: **A. Immediate**

Source in the file: Quizlet CEA201 Unit 13 multi — /https://quizlet.com/133385622/cea201-unit-13-multi-flash-cards

2 **In _____ addressing)EA = (A)(, the address field points to a memory location containing the effective address of the operand (used to implement pointers).**

A. Direct
B. Indirect ✓
C. Register
D. Immediate

Answer: **B. Indirect**

Source in the file: Quizlet Ch14 Addressing — https://quizlet.com/503607740/ch-14-instruction-sets-/addressing-modes-and-formats-flash-cards

3 **The advantages of _____ addressing (EA = R) are a small address field and no time-consuming memory references.**

A. Direct
B. Indirect
C. Register ✓
D. Displacement

Answer: **C. Register**

Source in the file: Quizlet CEA201 Unit 13 multi — /https://quizlet.com/133385622/cea201-unit-13-multi-flash-cards

4 **_____ addressing)EA = A + (R)(combines direct and register indirect capabilities and includes Relative, Base-register, and Indexing.**

A. Immediate
B. Displacement ✓
C. Stack
D. Implied

Answer: **B. Displacement**

Source in the file: Quizlet CEA-Chap13 — /https://quizlet.com/86502279/cea-chap13-flash-cards

5 **In _____ addressing)EA = A + (PC)(, the implicitly referenced register is the Program Counter (PC), used for branch instructions.**

A. Base-register
B. Relative ✓
C. Indexing
D. Register indirect

Answer: **B. Relative**

Source in the file: Studocu Ch14 Review — https://www.studocu.com/row/document/south-east-european-university/computer-architecture/chapter-xiv-instruction-sets-addressing-modes-and-formats/44778232

6 **In _____ addressing, the referenced register holds a main memory base address and the address field holds a displacement (used for relocatable code & recursive procedures).**

A. Indexing
B. Base-register ✓
C. Relative
D. Immediate

Answer: **B. Base-register**

Source in the file: Quizlet Ch14 Addressing — https://quizlet.com/503607740/ch-14-instruction-sets-/addressing-modes-and-formats-flash-cards

7 **_____ addressing operates on a LIFO queue where a register maintains the stack pointer (implied addressing on top of stack).**

A. Direct
B. Stack ✓
C. Relative
D. Base-register

Answer: **B. Stack**

Source in the file: Studocu Ch14 Review — https://www.studocu.com/row/document/south-east-european-university/computer-architecture/chapter-xiv-instruction-sets-addressing-modes-and-formats/44778232

8 **The value of the _____ field in the instruction format determines which addressing mode is to be used.**

A. Opcode
B. Mode ✓
C. Status
D. Parity

Answer: **B. Mode**

Source in the file: Quizlet CEA Unit 13 — /https://quizlet.com/86641740/cea-unit-13-flash-cards

9 **Which 6 interrelated factors determine the allocation of addressing bits?**

A. Number of addressing modes & Number of operands
B. Register vs memory & Number of register sets
C. Address range & Address granularity
D. All of the above ✓

Answer: **D. All of the above**

Source in the file: Quizlet CEA201 Unit 13 multi — /https://quizlet.com/133385622/cea201-unit-13-multi-flash-cards

Ch16: Processor Structure and Function

True / False — (2) صح / خطأ

1 **Inside the CPU, the four blocks of the ALU are Status Flags, Shifter, Complementer, and Arithmetic & Boolean Logic.**

Answer: **True**

Source in the file: Medium COA Part 2 — https://medium.com/@ckekula/computer-architecture-for-beginners-part-2-ae5015f34876

2 **Write After Read (WAR) is called antidependency, and Write After Write (WAW) is called output dependency.**

Answer: **True**

Source in the file: Quizlet Data Hazards — /https://quizlet.com/351965635/types-of-data-hazard-flash-cards

Multiple Choice — (11) اختيار من متعدد

1 **_____ registers enable the assembly programmer to minimize main memory references by optimizing register use.**

A. User-visible ✓
B. Control and status
C. MBR
D. I/OAR

Answer: **A. User-visible**

Source in the file: Quizlet OS Ch1 — /https://quizlet.com/226936749/os-chapter-1-flash-cards

2 **The _____ contains status flags: Sign, Zero, Carry, Equal, Overflow, and Interrupt Enable/Disable.**

A. MAR
B. Program Status Word (PSW) ✓
C. IBR
D. MQ

Answer: **B. Program Status Word (PSW)**

Source in the file: Medium COA Part 2 — https://medium.com/@ckekula/computer-architecture-for-beginners-part-2-ae5015f34876

3 **During the _____ cycle: PC -> MAR -> Address Bus -> Memory Read -> Data Bus -> MBR -> IR, and PC = PC + 1.**

A. Indirect
B. Interrupt
C. Fetch ✓
D. Execute

Answer: **C. Fetch**

Source in the file: Quizlet CEA201 Unit 2 — https://quizlet.com/159838675/cea201-unit-2-cea201-unit-4-cea201-unit-5-cea-201-unit-1-cea-201-unit-3-cea-201-unit-8-cea-201-unit-9-cea-201-unit-13-cea201-unit-7-cea201-unit-11-/cea201-unit-12-cea201-unit-14-c-flash-cards

4 **During the _____ cycle, the right-most N bits of MBR are transferred to MAR to read the effective address of the operand into MBR.**

A. Fetch
B. Indirect ✓
C. Interrupt
D. Halt

Answer: **B. Indirect**

Source in the file: WordPress COA VI — https://techiefood4u.wordpress.com/wp-content/uploads/2019/12/cao_vi.pdf

5 **During the _____ cycle, the PC is moved to MBR to be saved in memory (at a reserved address in MAR), and PC is loaded with the interrupt routine address.**

A. Fetch
B. Indirect
C. Interrupt ✓
D. Decode

Answer: **C. Interrupt**

Source in the file: WordPress COA VI — https://techiefood4u.wordpress.com/wp-content/uploads/2019/12/cao_vi.pdf

6 **A _____ hazard occurs when two or more instructions already in the pipeline need the same hardware resource at the same time.**

A. Data
B. Control
C. Resource (Structural) ✓
D. Branch

Answer: **C. Resource (Structural)**

Source in the file: Quizlet Adv Comp Arch — https://quizlet.com/552822799/advanced-computer-/architecture-exam-3-flash-cards

7 **Which Data Hazard is 'true dependency' (an instruction modifies a register/memory and the next instruction reads it before the write finishes)?**

A. Read After Write (RAW) ✓
B. Write After Read (WAR - antidependency)
C. Write After Write (WAW - output dependency)
D. RAR

Answer: **A. Read After Write (RAW)**

Source in the file: Quizlet Data Hazards — /https://quizlet.com/351965635/types-of-data-hazard-flash-cards

8 **A _____ hazard (branch hazard) occurs when the pipeline makes a wrong decision on a branch prediction and must discard fetched instructions.**

A. Resource
B. Data
C. Control ✓
D. Structural

Answer: **C. Control**

Source in the file: Vaia Pipeline Hazards — https://www.vaia.com/en-us/explanations/computer-science/computer-organisation-and-architecture/pipeline-/hazards

9 **Which branch technique uses a small, fast memory in the instruction fetch stage to store loop instructions without accessing main memory?**

A. Multiple streams
B. Loop buffer ✓
C. Delayed branch
D. Prefetch branch target

Answer: **B. Loop buffer**

Source in the file: Quizlet CEA201 Unit 14 multi — /https://quizlet.com/133385666/cea201-unit-14-multi-flash-cards

10 **Which branch technique fetches instructions from more than one possible branch path at the same time?**

A. Multiple streams ✓
B. Loop buffer
C. Delayed branch
D. Static branch

Answer: **A. Multiple streams**

Source in the file: Quizlet CEA201 Unit 14 multi — /https://quizlet.com/133385666/cea201-unit-14-multi-flash-cards

11 **In _____ branch, the processor always executes the instruction immediately after the branch (the delay slot) before the branch takes effect.**

A. Multiple streams
B. Prefetch branch target
C. Loop buffer
D. Delayed branch ✓

Answer: **D. Delayed branch**

Source in the file: Quizlet Pipeline — https://quizlet.com/108142741/computer-achitechure-pipeline-/flash-cards